



PREVENTING DRIVE BELT ALIGNMENT PROBLEMS

Dan Parsons
Plant Engineering
July, 1993

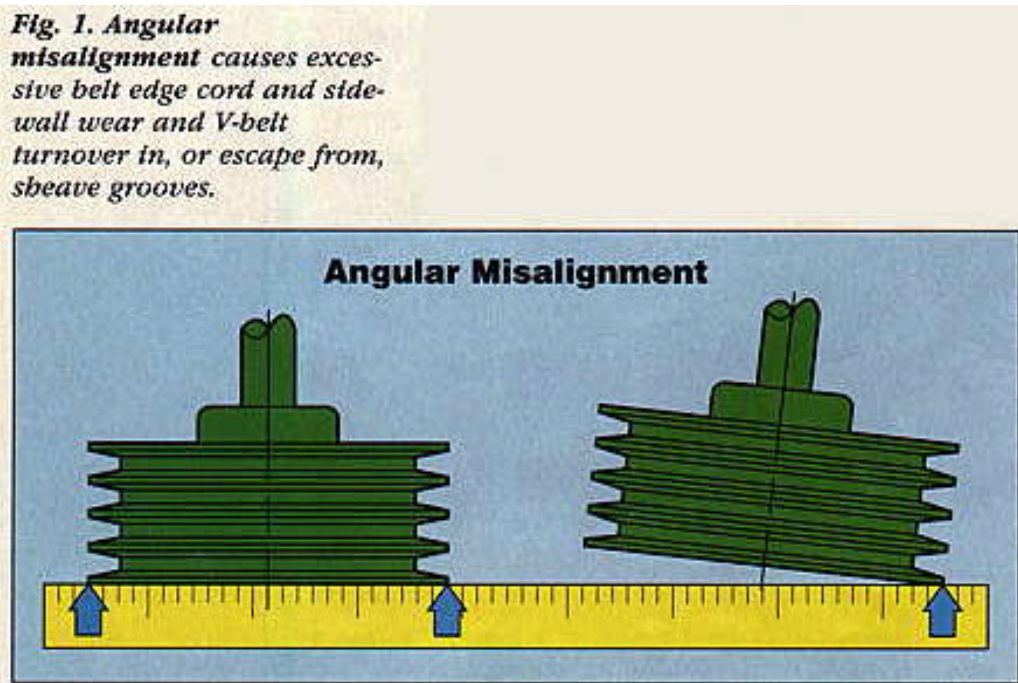
Amount of angular and parallel misalignment determines what action to take.

Misalignment is one of the most common causes of premature belt failure. The problem gradually reduces belt performance by increasing wear and fatigue. Depending on severity, misalignment can destroy a belt in a matter of hours or days.

While the basic forms of misalignment may be understood, accurate measurements and acceptable limits must be determined before corrective action is taken.

Types of Misalignment

Basically, any degree of misalignment, angular or parallel, decreases the normal service life of a belt drive. Angular misalignment (Fig. 1) results in accelerated belt/sheave wear and potential stability problems with individual V-belts. A related problem, uneven belt and cord loading, results in unequal load sharing with multiple belt drives and leads to premature failure.

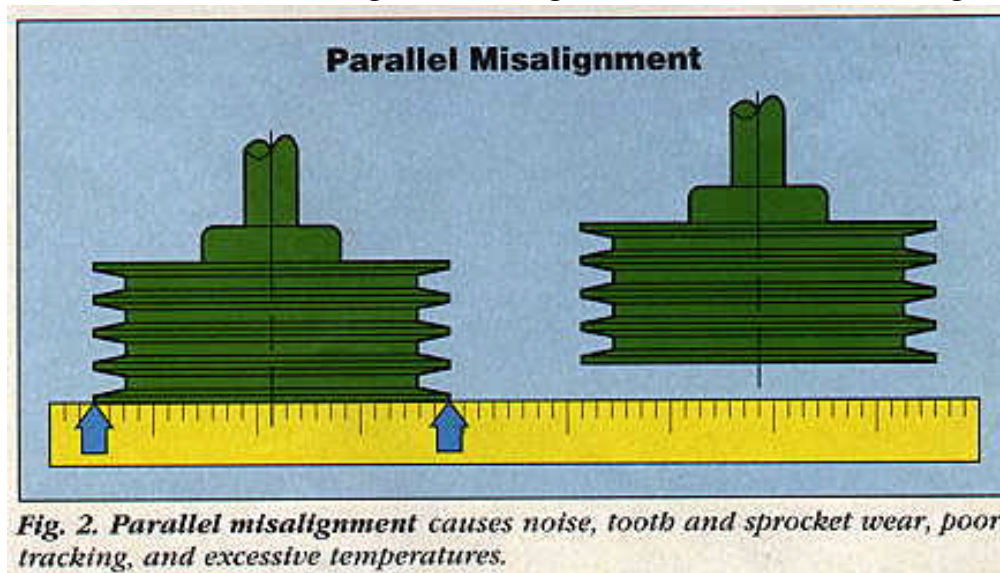




Angular misalignment has a severe effect on the performance of synchronous belt drives. Symptoms such as high belt tracking forces, uneven tooth/land wear, edge wear, high noise levels, and potential tensile failure due to uneven cord loading are possible. Also, wide belts are more sensitive to angular misalignment than narrow models.

Parallel misalignment (Fig. 2) also results in accelerated belt/sheave wear and potential stability problems with

individual belts. Uneven belt and cord loading is not as significant a concern as with angular misalignment.



However, parallel misalignment is typically more of a concern with V-belts than with synchronous belts. V-belts run in fixed grooves and cannot free float between flanges to a limited degree as synchronous belts can.

Parallel misalignment is generally not a critical concern with synchronous belt drives as long as the belt is not trapped or pinched between opposite flanges and the belt tracks completely on both sprockets.

Synchronous belt sprockets are designed with face widths greater than belt widths to prevent problems associated with tolerance accumulation, and to allow for a small amount (fractions of an inch) of mounting offset.

As long as the width between opposite sprocket flanges exceeds the belt width, the belt automatically aligns itself properly as it seeks a comfortable operating position on both sprockets. It is normal for a synchronous belt to lightly contact at least one sprocket flange in the system while operating.

Measuring Misalignment



The most common tools for measuring misalignment are a straightedge and string. The improper use of either tool, especially a string, can result in erroneous conclusions (Fig. 3).

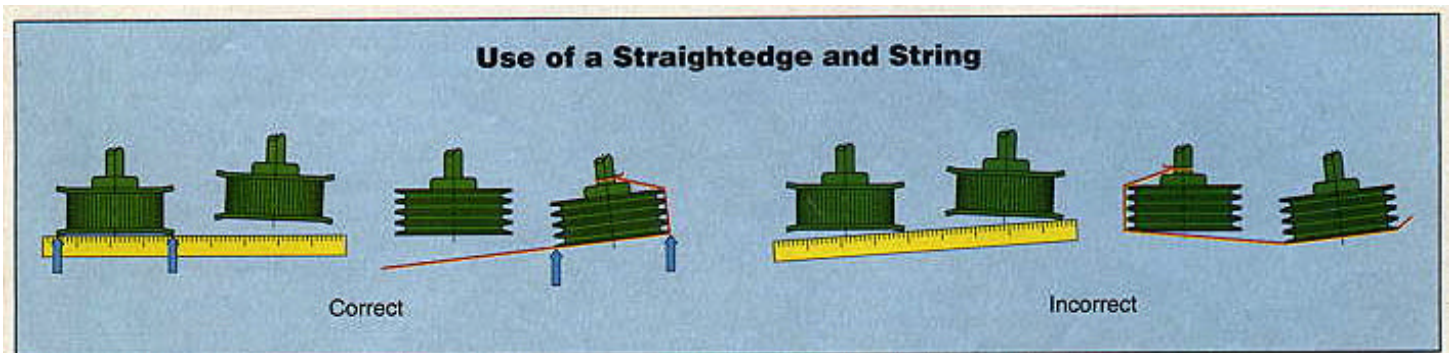


Fig. 3. Correct and incorrect ways to use a straightedge and string to check for misalignment are shown.

A straightedge should be used to project the orientation of one sheave or sprocket face with respect to the other. Orientation is also accomplished with a string, as long as it remains straight without any kinks or breaks.

When preparing to measure parallel misalignment, verify that edges of both sheaves or sprockets are of equal thickness, or quantify the difference in thickness. Align sheave grooves or sprocket faces directly with respect to one another, rather than the outside surfaces of the sheaves or sprockets. It may be necessary to mount sheaves or sprockets with the outside surfaces offset with respect to one another in order to properly align grooves or surfaces on which belts operate.

Sprocket flanges should be inspected to be sure they run true. A bent flange could result in erroneous measurements if the straightedge or string rests against the outside edge of a damaged flange during the inspection process.

To determine how much misalignment is acceptable and at what point it becomes excessive, alignment must be measured, quantified, and compared to the belt manufacturer's recommendations for particular types of drives. These recommendations are found in drive design manuals.

Quantifying Misalignment

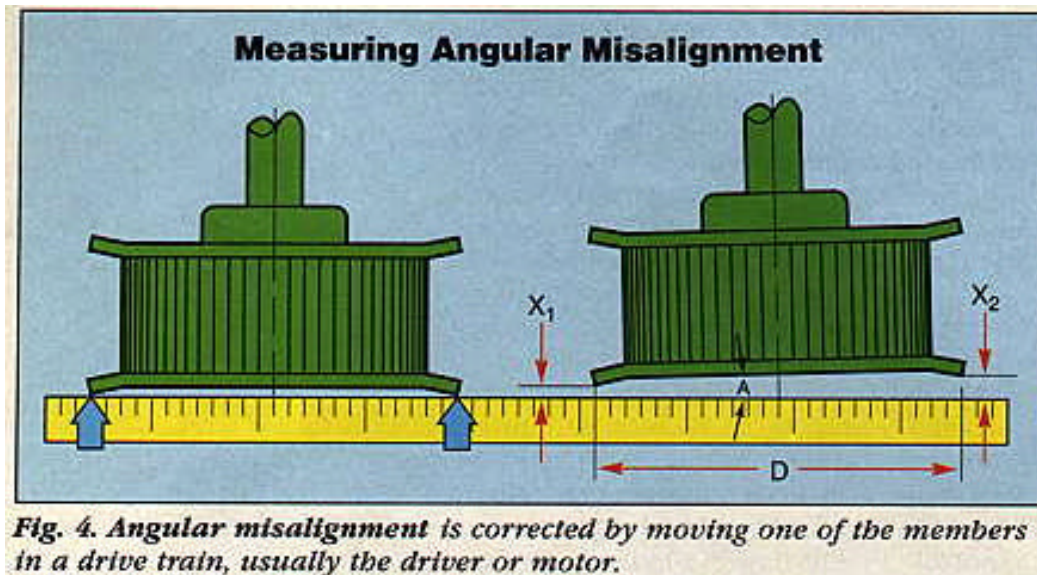
Misalignment is quantified mathematically or compared to some general rules of thumb for quick and easy results.

Angular misalignment is quantified into a real value by taking measurements (Fig. 4).



GatesFacts™ Technical Information Library
Gates Compass™ Power Transmission CD-ROM version 1.2

The Gates Rubber Company
Denver, Colorado USA



The actual angle of misalignment is defined by the difference in clearance between the straightedge or string and the outside surface of the sheave or sprocket across the diameter. The mathematical relationship is:

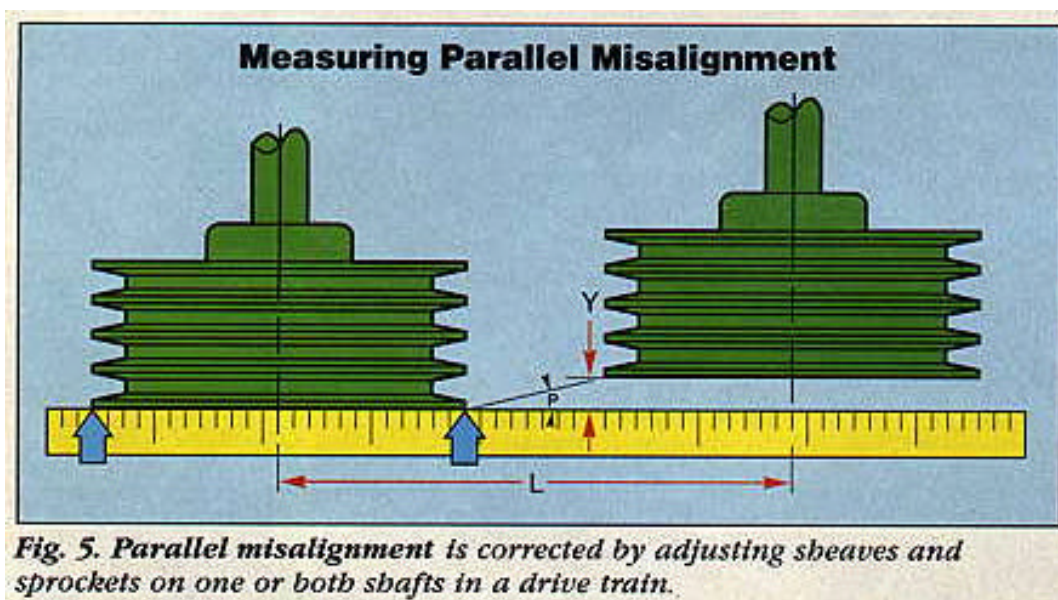
$$A = \text{ArcTan} [(X_2 - X_1)/D]$$

where

A = angular misalignment, deg.

D = diameter of sheave or sprocket, in.

X = distance from straight edge to sheave flange, in.





The angle of parallel misalignment is defined by the difference in clearance between the straightedge or string, and the outer surfaces of the two sheaves or sprockets across the span length of the belt (Fig. 5).

The mathematical relationship is:

$$P = \text{ArcTan} (Y/L)$$

where

P = parallel misalignment, deg.

Y = distance from straightedge to sheaves, in.

L = center distance between sheaves, in.

The total allowable misalignment recommended for V-belts is 1/2 deg. While individual V-belts are capable of handling misalignment up to 6 deg before becoming unstable, maintaining the misalignment to within 1/2 deg maximizes belt life. Joined V-belts tolerate misalignment up to 3 deg before significant tieband damage occurs.

The total amount of misalignment recommended for synchronous, 60-deg, and V-ribbed belts is 1/4 deg. These drives are less tolerant of misalignment than conventional V-belt drives, and must be installed accurately.

When determining if a V-type drive system is aligned within these recommendations, angular and parallel misalignment must be measured, quantified, and added together. The total sum of angular and parallel misalignment is compared to the belt manufacturer's recommendations for the particular type of drive.

Since synchronous belts are particularly sensitive to being pinched or trapped between opposite sprocket flanges, sprockets must be installed so there is clearance between the belt and both flanges. This installation eliminates parallel misalignment, and does not have to be quantified and added to angular misalignment for a total value.

Rules of Thumb

Maintenance technicians may not find it practical or possible to accurately calculate total misalignment in a system while determining if it is in acceptable alignment. It is also difficult to visualize small fractions of an angle such as 1/4 or 1/2 deg. These angles are illustrated with the following rules of thumb:

- For V-belt drives: 1/2 deg = approximately 1/10-in. offset per foot.
- For synchronous, 60-deg angle, and V-ribbed drives: 1/4 deg = approximately 1/16-in offset per foot.

These rules are used to estimate the amount of angular and parallel misalignment visually rather than by calculating numerical values.

Tips for Aligning Drives

Dual plane drive alignment. The processes described above permit alignment checking in one plane only. Shafts may be misaligned in either of two different planes, or both. For example, a drive with horizontal shafts is aligned in one plane using the techniques described above, then lined up in the second plane using a bubble level. The bubble level is used to see that both shafts are parallel with respect to the ground. If a drive has vertical shafts, the bubble level is used to make certain both shafts are perpendicular to the ground.



GatesFacts™ Technical Information Library

Gates Compass™ Power Transmission CD-ROM version 1.2

The Gates Rubber Company
Denver, Colorado USA



Parallel alignment. Parallel misalignment is difficult to determine since an accurate common reference plane is not always available. Synchronous belt drives are checked by making sure the belt is not pinched between opposite flanges or does not track off any unflanged sprockets. If the shafts are horizontal, and one is located vertically above the other, a plumb bob or bubble level is used to determine if the sheaves or sprockets are in line with each other. A single V-belt could also be hung in an outside sheave groove from the upper shaft to indicate the proper position of the lower sheave.

Angular alignment. After alignment and tension of a synchronous belt drive are set as accurately as possible, a simple test is used to make sure the system is lined up properly.

Carefully turn the drive over by hand and observe which direction and how fast the belt tracks toward one flange. The belt should move slowly enough that several revolutions are required for the belt to move from one sprocket flange to the other. The more rotations required, the better the system is aligned.

The drive should then be stopped and the rotation reversed. The belt should track in the opposite direction at about the same speed as before. If the belt continues to track in the same direction as before, the system is angularly misaligned and needs adjustment.

When adjusting angular alignment of synchronous belt drives, it is helpful to know what effect small adjustments have on the direction of belt tracking. Since synchronous belts tend to track "downhill" to a state of lower tension, or toward a shorter center distance, this fact determines the direction shafts need not be adjusted.

Related components, such as brackets and platforms, should also be checked for proper design and placement. These parts must be strong enough to withstand peak forces exerted by drives without bending or flexing.



Installation And Storage Of Synchronous Belt Drives

PA NOTE

This bulletin describes the correct procedures for installing, storing and maintaining all types of synchronous belts. The basic steps for installing a synchronous belt drive are the same, regardless of belt type or size. Gates catalogs 17595 (Poly Chain® GT® Drives), 17190 (PowerGrip® HTD® Systems for Industrial Drives), and 17181 (PowerGrip® Timing Belt Systems for Industrial Drives) include detailed information on properly designing synchronous belt drives.[Also new PowerGrip Belt Systems, 17195.]

Installing the Sprockets

Before installing the hardware, check the shafts and tapered surfaces of the bushing and sprocket for contaminants. Any rust, dirt, oil, or grease should be removed from the mating surfaces. Lubricants should never be used when installing the sprockets or pulleys. If a lubricant is used, bursting pressures may be generated, resulting in damage to the hub of the sprocket.

With the key installed in its keyseat, slide the bushing into place on the shaft. If the bushing is difficult to install on the shaft, a screwdriver blade can be carefully wedged into the bushing's saw cut. If a bored-to-size sprocket is used, it can be installed on the shaft with the keyway in place.

With the bushing on the shaft, loosely insert the bolts through the sprocket and into the bushing flange. The sprocket may be reverse mounted by inserting the bolts through the bushing and into the hub (QD bushing sizes JA through J only). Do not use any type of lubricant on the bolts, bushings or sprocket.

With the sprocket and bushing loosely assembled, initially align the assemblies.

The bushing bolts can now be tightened evenly and progressively around the bolt circle. Tighten the bolts using a torque wrench to the recommended torque values for the appropriate bushing size. It is important to use a torque wrench to insure that the sprocket is seated properly on the bushing, which will help minimize the wobble and eccentricity of the sprocket. The recommended values are listed in the table on next page.

Bushing	Bolts(in)	Torque
---------	-----------	--------



GatesFacts™ Technical Information Library

Gates Compass™ Power Transmission CD-ROM version 1.2

The Gates Rubber Company
Denver, Colorado USA



Style	Qty	Size	Wrench Force(ft-lbs)
QT	2	1/4 x 3/4	6
JA	3	10-24 x 1	5
SH &SHDS	3	1/4 x 1 3/8	9
SD	3	1/4-20 x 1 7/8	9
SK	3	5/16-18 x 2	15
SF	3	3/8-16 x 2	30
E	3	1/2-13 x 2 3/4	60
F	3	9/16-12 x 3 3/8	75
J	3	5/8-11 x 4 1/2	135
M	4	3/4-10 x 8 3/4	225
N	4	7/8-9 x 8	300
P	4	1-8 x 9 1/2	450
W	4	1 1/8 x 11 1/2	600
S	5	1 1/4-7 x 15 1/2	750

Note that there should be at least a 1/8" to 1/4" gap between the sprocket hub and the bushing flange. This is normal under recommended installed conditions. Do not attempt to tighten the bolts to eliminate this gap.

If a bored-to-size sprocket with a conventional rectangular key is used, it can be installed on the shaft with the key in place. Be sure to install and tighten the set screws.

Depending on the bore size, some Poly Chain® GT® bored-to-size sprockets may use a Gib key rather than a conventional rectangular key. A Gib key differs in shape and installation procedures from a conventional key. A Gib key has a slight taper on one surface. When the key is inserted, the taper creates a wedging action that clamps the sprocket to the shaft. [Gib keys were eliminated in the 1980's.]

Every bored-to-size Poly Chain® GT® sprocket that uses a Gib key comes with an installation kit. The kit contains a packet of "Never Seez" (TM Bostik, Inc.) (noncorrosive) material and two color coded shims. The first step when installing a Gib key is to apply the Never Seez material to the entire key and insert the key into the keyway of the shaft/sprocket assembly. The key should be lightly tapped until firmly seated. If the gap between the hub face and the key head is less than 1/16", remove the Gib key and place the shim stock between the key and shaft keyway. Reinstall the key as above, checking the clearance between the key head and hub face. It may be necessary to



repeat the above steps, adding shim material until the gap is more than 1/16" when the key is firmly seated. Once the sprocket is secured, the set screw can be tightened.

When installing a Gib key, avoid using excessive force to seat the key. If too much force is applied to the key, the sprocket hub may be split.

Recheck Alignment

The alignment should be checked once again. Remember that the maximum recommended misalignment for a synchronous drive is $\frac{1}{4}$. A suitable straight edge should be used. When using a straight edge, check for two points of contact on each sprocket. This will easily identify parallel misalignment. Check for both angular and parallel misalignment. See Figure 1 on page 3.

Angular misalignment can be more difficult to identify, but can be done by checking alignment in both directions. First, place the straight edge across the face of the driveR and check the alignment of the driveN. Then place the straight edge across the face of the driveN and check the driveR alignment. This procedure will identify both parallel and angular misalignment. If the alignment appears to be incorrect, loosen the bolts and adjust the sprocket assembly's position. This procedure may have to be repeated to achieve the desired alignment. At this point, complete the sprocket installation by a final tightening of the bolts for a bushed sprocket or of the set screws for a bored to size sprocket.

Installing and Tensioning the Belt

Now that the sprockets are in position, the belt can be installed. Adjust the center distance or idler position so the belt can be easily slipped over the sprockets or pulleys. The belt should never be pried onto the drive. If the belt is pried onto the drive, the tensile member could incur unseen damage, resulting in premature belt failure. With the belt installed on the drive, take up the belt length with either shaft adjustment or an idler.

The proper belt tensioning values can be calculated by using the appropriate drive design catalog. Once the required tension values are known, begin by tightening the drive to an approximate initial tension. Measure the belt tension using either a pencil type tension tester or a spring scale. Compare the force required to deflect the belt 1/64" per inch of span to the recommended belt tension. If the force is higher than recommended, loosen the drive. If the force is lower than the recommended tensioning force, tighten the drive. Continue to adjust and check until the measured force falls between the minimum and maximum recommended tensioning forces.

Any drive guards or machine components should now be reinstalled. At this point, the drive should be ready for operation. Run in the drive briefly, carefully looking and listening for any unusual signs.

Belt Storage



GatesFacts™ Technical Information Library

Gates Compass™ Power Transmission CD-ROM version 1.2

The Gates Rubber Company
Denver, Colorado USA



To achieve maximum belt performance, proper belt storage procedures should always be practiced. If the belts are not properly stored, their performance can be adversely affected.

Before we discuss proper storage procedures, let's examine some methods of storage to avoid.

- 1) Do not store belts on floors unless they are protected by the appropriate packaging.
- 2) Do not store near windows where the belts may be exposed to direct sunlight or moisture.
- 3) Do not store near electrical devices that may generate ozone (transformers, electric motors, etc.).
- 4) Do not store in areas where solvents or chemicals may be present in the atmosphere.

Synchronous belts up to 120 inches are normally stored in a "nested" configuration. "Nests" are formed by laying a belt on its side and placing as many belts inside the first belt as possible without undue force. The nests can be stacked when they are tight and rotated 180° from the nest below.

Synchronous belts over 120 inches long may be rolled up for storage and shipment. The rolls can then be stacked for storing purposes. Small bend radii which could damage the belts should be avoided.

Poly Chain GT belts are individually boxed which makes storage on shelves very convenient. This also protects the belt.

Per RMA Bulletin No. IP-3-4/1997, when proper storage procedures are followed, and the belt is stored at temperatures less than 85 °F and relative humidities below 70%, the belt's quality should not significantly change within 8 years. If the storage temperature exceeds 85 °F, the storage limit for normal belt service levels will be significantly reduced.

Preventive Maintenance

An excellent publication, "Belt Preventive Maintenance Manual", form #14995, is available from Gates. It includes detailed discussion of many subjects, including failure analysis and troubleshooting problem drives.



GatesFacts™ Technical Information Library

Gates Compass™ Power Transmission CD-ROM version 1.2

The Gates Rubber Company
Denver, Colorado USA



Misalignment

Sheave misalignment on V-belt drives should not exceed $\frac{1}{2}$ or $\frac{1}{10}$ " per foot of centers. For micro-v or synchronous belts it should not exceed $\frac{1}{4}$ " or $\frac{1}{16}$ " per foot of centers. For V-belts the generally accepted maximum misalignment before stability (turnover) becomes a serious concern is 6° . Obviously other factors can also influence stability such as insufficient tension, worn grooves, and shock loading. We also need to note that within the $\frac{1}{2}$ for V-belts we would expect no significant impact on belt fatigue life. At 6° misalignment, not only is belt stability an issue, but stress fatigue to the belt would be very significant.



Synchronous Belt to Pulley Fit During Installation

Historically the installation of trapezoidal belts presented very few problems over the years. The 40 - 50° angle on the belt teeth and pulley grooves allowed relative ease of belt installation regardless of pitch or pulley diameter. Belt tooth to pulley groove clearance was very good (minimal) providing a belt with excellent backlash characteristics.

With the introduction of the round HTD® tooth form in the early 1970's, some problems became evident. The ability to hand-wrap longer belts around larger diameters with no tension became more difficult. This was compounded if the sprocket diameter tolerance was too great and/or the belt pitch was too short. Even the greater clearance of the HTD® system did not always help, although it made for poorer positioning accuracy.

As the power density of today's newer, modified curvilinear profiles increased even further, the belt development engineer's problems of matching the ideal belt construction to the sprocket groove clearance became a greater concern. The modified profile belts did however allow for improved backlash through minimized clearance values - a plus on positioning applications. However, tolerance stack-up can again create a situation where user cannot hand-wrap a belt more than 90° or so around a larger sprocket.

Switching from a fiberglass tensile belt to an aramid tensile belt can create yet other problems due to tendency for aramid to shrink slightly in a high moisture (humidity) environment.

In the above situations it may become necessary to work a little harder at installing the belt. Gates has successfully used the following process for years. Completely wrap the belt into the small sprocket's grooves (generally the driver). Hand-wrap the belt in the large sprocket as far as possible without undue force. Alternately add tension (typically through the motor base jackscrews) and rotate the drive by hand. Each time you do this one or more additional belt teeth will seat in the grooves. Repeat until the belt is completely seated in the large sprocket. Check the belt for the proper recommended tension and lock the motor base in place.

If the above process fails to do the job, it is quite possible that either the large sprocket is way out of tolerance on the high end and/or the belt is too short. In this instance you will not be able to pull the belt into pitch with the sprocket. Either or both parts will need to be replaced.



Synchronous and V-Belt Drive Tensioning PA NOTE

Static Tension

What is the best tension for a V-belt drive? You might ask this question when the belt is on the verge of slipping under the highest possible load. This slipping point may not be easily found and does not allow for changes in belt growth, wear or even environmental conditions such as humidity. Since synchronous belts cannot slip on a sprocket, the best tension is the minimum tension that will keep the belt's teeth properly seated in the grooves under peak loads.

With too little tension in a V-belt drive, slippage can occur and lead to spin burns, cover wear, overheating of the belt, and even overheating of the bearings. Not enough tension in a synchronous drive will cause premature tooth wear or possible ratcheting (jumping teeth) that will destroy the belt or possible shaft breakage. Too much tension overloads bearings and bracketry and can lead to early bearing failure. Overtensioning will also shorten belt life.

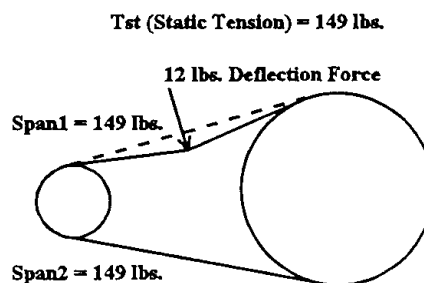


Figure A

Total tension required in a belt drive depends on the type of belt, the design horsepower (horsepower x service factor) and the RPM of the drive. Since running tensions cannot generally be measured, it is necessary to tension a drive to a static tension (a stationary installation tension). The most common method of establishing a static tension is by the force/deflection method. As shown in Fig. A, once a calculated force is applied in the center of a span with a known deflection, the recommended static tension will be established. Most design catalogs will provide force and deflection formulas. Gates computer design programs can also provide these calculations.

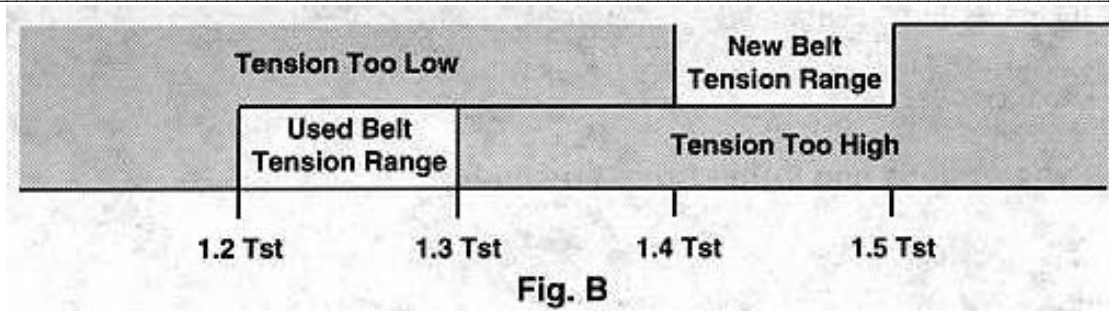
When installing new synchronous or V-belts for the first time (or after a belt has been removed from a drive), installation tensions should be set higher. Generally 1.4 to 1.5 times the normal static tension. This higher static tension is necessary because drive tension drops rapidly during the seating in process. This extra initial tension does not adversely affect bearings because it decays so rapidly. For maintenance retensioning of existing drives, tension can be 1.2 to 1.3 times the normal static tension as shown in Fig. B.



GatesFacts™ Technical Information Library

Gates Compass™ Power Transmission CD-ROM version 1.2

The Gates Rubber Company
Denver, Colorado USA

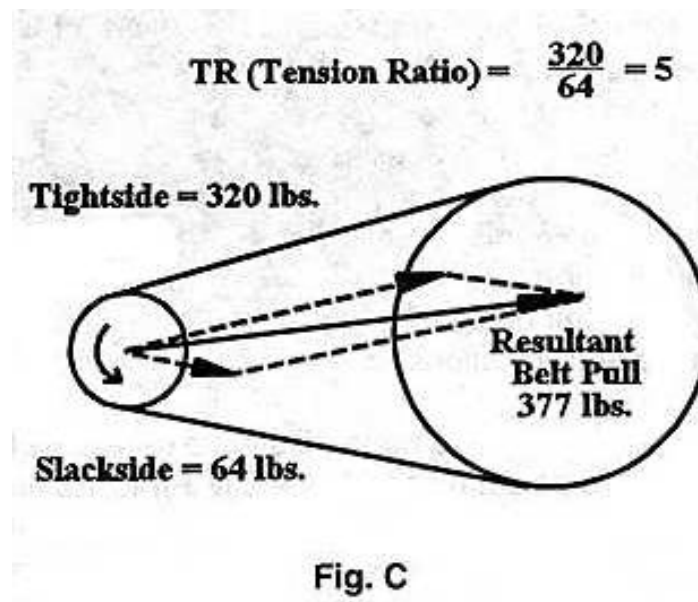


It has always been recognized that we must give the user some tolerance on tension (since it is difficult to install to an exact value), so we have used the idea of a range. The closer a belt drive can be installed to the ideal 1.0 tension value, the lower the stress on all drive components.

These calculations are per belt, strand or rib. If multiple belts/ribs are used, the static tension and deflection forces should be multiplied by the number of belts/ribs when tensioning the entire drive. It is not necessary to multiply the deflection distance by the number of belts/ribs.

Belt Pull

The preceding discussion dealt with static tensions only. For a two-point stationary drive the span tensions are shown in Fig. A. Once a drive starts operating, different tensions will develop in a drive. Usually most drives operate near a 5:1 tensioning ratio as show in Fig. C. It is under these dynamic tensions (working tensions) that a force is imposed on the bearing (belt pull) that is much greater than the static force belt pull. A vector analysis of these dynamic tensions is shown in Fig. C.





Installation Procedure for Synchronous Sprockets & Belts

PA NOTE

With the use of many high capacity synchronous belt drives in the Power Transmission Industry there are installation, alignment, tension and belt tracking issues to address. Maintenance personnel need to be aware of these drive characteristics for a successful installation or conversion. Although similar to V-belts, synchronous belt drives will require some greater attention to detail in order to optimize performance.

IMPORTANT: *Insure all lock out and tag procedures are implemented prior to any belt or drive inspection, maintenance or installation!*

‘QD’ BUSHED SPROCKET MOUNTING

1. To install the sprocket, first insure all parts are clean and devoid of any lubricants, coatings, or contaminants. Slide the bushing on the shaft, install the key and **hand tighten** the setscrew over the key. **DO NOT USE EXCESSIVE FORCE!**

Do not use lubricants or antiseize products on the sprocket taper bore or the bushing taper surface, as these products may cause excessive component forces using recommended cap screw torque values. The result could be cracked and broken sprocket hubs or bushings.

2. For a *conventional mount* slide the large end of the sprocket taper bore into position over the mating tapered bushing surface, aligning the unthreaded holes in the sprocket hub with the threaded holes in the flange of the bushing. Hand tighten the cap screws with lock washers installed.

Due to sprocket design or clearance on a particular drive, some sprocket assemblies will allow a *reverse mount* procedure by reversing the entire sprocket-bushing combination. This results in the bushing flange facing out but still allows the cap screws to be installed from the outside of the assembly; the cap screws are placed through the unthreaded holes of the bushing flange and into the threaded holes of the sprocket hub. These are typically threaded partially due to the long length of the hub.

3. Torque the bushing cap screws alternately and progressively. **DO NOT USE EXTENSIONS** on the torque wrench handle. **Do not use impact wrenches!** These procedures will prevent cracked or broken bushings and sprocket hubs. Finally, tighten the keyway set screw using a standard allen wrench. A gap should be present between the face of the sprocket hub and the face of the bushing flange to insure adequate grip on the shaft. No gap typically indicates a bore or shaft misfit.

Alternately and progressively tightening the cap screws is extremely important when installing a synchronous sprocket. A synchronous drive is not as forgiving as a V-belt drive and performance can be more adversely affected with a cocked bushing/sprocket.



DRIVE ALIGNMENT

4. After sprockets are installed, place the belt on the drive, take-up the slack, and check for proper alignment. Using a straight-edge or metal tape for long centers, you can easily check for misalignment. Heavy string can be used with care on short center distance drives.

5. Line the straight-edge along the face of both sprockets, checking the driveR and driveN sprockets separately. Misalignment will show up as a gap between the outside face and the straight-edge. Although recommended drive alignment allows for 1/16 inch misalignment per foot of center distance which equals 1/4 degree for angular or parallel values, try to keep any misalignment to a minimum to allow for optimum tooth engagement and belt performance. If flanges are damaged, missing or cause interference with alignment procedures, use 1/4 inch thick, flat ceramic magnets on the sprocket face for a helpful tool. They will provide a flat alignment surface, extend beyond the flange and are available from any hobby shop or hardware store.

BELT TENSION

6. Belt tensioning is the next step and the objective is to apply the least amount of static tension to handle actual belt load. Installation tension can be measured with a tension tester in conjunction with the appropriate deflection value and forces. This procedure will provide adequate static tension to prevent the belt teeth from ratcheting or jumping sprocket grooves under load. **TOO MUCH OR TOO LITTLE TENSION REDUCES DRIVE COMPONENT LIFE.**

7. Using the force deflection method of tensioning choose an accessible span and identify the span length from where the belt exits one sprocket and enters the other. The drive belt should be deflected midspan 1/64 inch of deflection for each inch of span length between adjacent shafts or pulleys. The range of force recommended to deflect the belt the appropriate distance is available from the belt manufacturer (contact Gates Product Application HELPLINE @ 303/744-5800 with drive details). Or, the range of force can be calculated for the exact drive from tension formulas in the Gates Drive Design Catalogs or the Design Flex™ II computer program.

8. Using a deflection tension tester, place a straight-edge on the top of the belt so it rests on both sprockets. The belt must now be deflected midspan by the distance and force predetermined from Step 7.

The lower portion of a single shaft pencil tension tester has an o-ring which can be adjusted to fractional inch increments and should be set at 1/64 inch per inch of span length, or if span distance is 48 inches, the o-ring should be set at 48/64 or 3/4 inches.

The upper portion of a tester also has an o-ring which should be set to zero. When you apply the force equally to the belt midspan and the bottom o-ring reaches the same level as the reference surface of the straight edge, the top o-ring should have moved indicating the pounds force of deflection. You now compare this value with the value(s) from Step 7. On multiple shaft pencil tensioners, use the sum of all values to determine the total deflection force.



The single shaft tensioner allows a maximum deflection of 30 pounds, while the double barrel tensioner allows a maximum deflection of 66 pounds.

Beyond the 66 pound values, one may need to use a heavy duty spring scale (multiple scales for large deflection forces) or Gates Sonic Tension Meter which can measure actual static tension. A length of chain and short section of pipe used with the spring scale will help equalize deflection forces across the belt width.

If the measured deflection force reads lower than desired, the drive must be tightened; and if the value reads higher than desired, the drive must be loosened. Also remember lower tensions are the cause of belt ratcheting while higher tensions are also imposed on shafts, bearings and the drive structure. With tensioning at the proper value recheck the sprocket alignment and adjust or proceed as necessary. Belt manufacturers do recommend new drives or belts be tensioned at values approximately 25% higher than typical maintenance or used belt values to allow the new belt to seat in. This greater initial tension decreases quickly under load on a new belt due to seat in and should not seriously affect the belt, bearings and other components.

BELT TRACKING

9. Due to the manufacturing process of synchronous belts there is a normal tendency for the belt to track toward a flange. This can frustrate many people trying to track the belt in the middle of the sprockets. The sprocket is not crowned as in older flat belt pulley designs or conveyor applications.

The normal tendency of the belt is to track one direction or the other. If you are comfortable with sprocket alignment as explained above, this condition should not pose a problem. A belt that tracks immediately to a flange when the drive is hand rotated indicates alignment is good. Be careful, if hand rotation is possible, to keep away from rotating parts and prevent loose items from getting caught in drive components.

10. The static condition alignment procedure may not address drive alignment in a dynamic mode, particularly when loads on drive components can be variable and extreme. This load-induced misalignment can cause belts to track improperly and reduce belt tensions. Structural reinforcement of the drive framework may be required to address dynamic load drive misalignment. This condition should be investigated and addressed immediately upon discovery to prevent drive performance problems or premature drive failure.